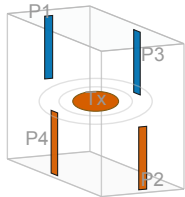


Calibrated posterior pipeline for RF capsule localization

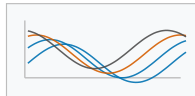
full-wave simulation evidence now; physical phantom validation is the next gate

A Four-receiver RF problem



four receivers on vertical
side walls
green: upper pair;
orange: lower pair
top and floor faces
have no receiver

B Full-wave data generation



S-parameter sweep $\rightarrow$
180-D RF features



N=3935 configurations
position-grouped 5-fold CV

C Conditional posterior

condition c

RF measurement;
fixed geometry

learned conditional transport



source

noise z_0 ,
 $t=0$

posterior

samples $p(x | c)$,
 $t=1$

D Evidence and boundary

Point accuracy

CFM 0.354 cm;
backbone 0.340 cm

Calibration

cov90 = 0.970;
cov95 = 0.975

Discrimination

error-spread rank
 ρ approx 0.60

Modality

bimodal only under
geometric ambiguity

Physical phantom + SDR
validation
future gate;
not claimed here